SARAH HALEVY

AI ENGINEER | BACKEND & AI SYSTEMS

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PROFILE

AI Engineer with proven end-to-end ownership across production web applications, backend systems, distributed data platforms, databases, and AI-driven products. Sole engineer of a live production data-governance platform, designing its PostgreSQL architecture, row-level security, audit trails, trust-scored data modeling, human review workflows, AI translation automation, and a bilingual offline-first PWA. Works with AI tools daily as an integral part of the development workflow, with a strong emphasis on Claude.

PROFESSIONAL EXPERIENCE

AI Engineer | Ainsof Capital

2026 | Remote

- Sole engineer of a live production data-governance platform, owning it end to end, from architecture design through launch.
- Designed a claims-based data model where every fact is sourced, trust-scored, auditable, conflict-detected, and routed through human review when needed.
- Built the PostgreSQL backbone across 108 SQL migrations, 3 schemas, 31 RLS policies, triggers, RPC functions, optimistic concurrency, and idempotency keys.
- Improved API latency from 582ms to 40ms through SQL optimization, backend tuning, and improved data access patterns.
- Built a human-gated AI translation pipeline using the Claude API, ensuring machine output never overwrites human edits.
- Delivered a 28-screen governance center and bilingual offline-first travel PWA with Hebrew RTL, offline maps, and REST APIs.

Data / ML Engineering Bootcamp | VAST Data

2025 | Internship

- Developed an end-to-end weed detection pipeline using Python, PyTorch, and MobileNetV3, including image retrieval from MinIO, inference execution, and PostgreSQL result storage.
- Built an image embedding system using CLIP/ViT-mini, publishing vectors to Kafka and loading them into pgvector for similarity search and analytics.
- Containerized backend and AI services in a scalable Docker-based environment to support reproducible development and deployment workflows.

SELECTED PROJECTS

Plant Species & Health Classification

Computer Vision | YOLO | CLIP | FastAPI | AWS EC2 GPU

- Built an end-to-end computer vision pipeline using YOLO and CLIP for plant detection, classification, and health analysis.
- Developed a GPU-accelerated FastAPI backend optimized for low-latency inference and deployed the system on AWS EC2.

Context-Aware AI Git Server

Python | MCP Server | AI Agents | Safe Automation

- Developed a production-grade MCP server enabling AI agents to analyze Git state, repository history, and code context.
- Implemented SafeGitRunner and a human-in-the-loop approval flow to enforce deterministic execution, command safety, and secure automation.

TECHNICAL SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, C#, C++, SQL

Backend: Node.js, .NET Core, FastAPI, Express, REST APIs, Microservices, Serverless / Edge Functions

Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, Responsive UI, RTL / i18n, PWA

Databases & Data: PostgreSQL, Supabase, RLS, SQL Server, MySQL, MongoDB, DynamoDB, MinIO, pgvector

Cloud & DevOps: AWS EC2, AWS S3, Docker, Kubernetes, GitHub Actions, CI/CD, Linux, Git, GitHub

AI / ML: PyTorch, Claude API, LLM Integration, Computer Vision, YOLO, CLIP, MobileNetV3, ViT-mini, CNNs

AI Tools: Daily AI-assisted development across the full workflow, primarily with Claude

Testing & Tools: Vitest, Playwright, PyTest, Unit & Integration Testing, Postman, Swagger, VS Code, Visual Studio

Concepts: OOP, Clean Architecture, Design Patterns, SQL Optimization, Distributed Systems, Agile

EDUCATION

KamaTech - UltraCode AI Program

Computer Science, Artificial Intelligence & Systems Engineering | Sep 2023 - Nov 2025 | Grade: 98

MAHAT - Government Institute of Technology & Science

Practical Engineering Diploma, Computer Software Engineering | Sep 2023 - Jul 2025 | Grade: 97

LANGUAGES

Hebrew: Native

English: Fluent